Text-to-SQL Models Make Choices Without Saying So [Experiment, Analysis & Benchmark]: Supplementary Material

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This supplement complements the paper; it does not repeat it. Section, figure, table and equation numbers without an S refer to the paper. The data and code behind every number are in the reproducibility package of the repository.

A QUERY GENERATION DETAILS

Algorithm S1 lists the construction of a request set. Figure S1 shows, for each feature, the share of 152 databases whose schema allows it and the share where a generated query with it was accepted by both model reviewers; and Table S1 the queries delivered on the 12 evaluation databases.

Formal statement. Let $\mathcal{L}(K_D)$ be the queries that the catalogue K_D of database D and the type rules allow, $\delta(p)$ a query's arithmetic and step depth, and $O(p)$ its operations. For target depths d and required operations R the generator samples from

$$\mathcal{P}(D, d, R) = \{p \in \mathcal{L}(K_D) : \delta(p) = d,\ R \subseteq O(p),\ I_D(p) = 1\}, \quad (1)$$

where $I_D(p) = 1$ means that all data checks pass. It keeps a partial query z only if some completion can still reach the target,

$$\mathrm{can}(z, d, R) = \mathbf{1}\{\exists (d, U) \in \mathrm{Reach}(z) : R \subseteq U\}, \quad (2)$$

where $\mathrm{Reach}(z)$ lists the depths and operation sets still possible. The look-ahead is exact for the rules, and no query is drawn twice. For a request set F with specific requests q_A, q_B and open request $q_\circ$, let $H_{ij}^{(r)} = 1$ when reviewer r accepts request i for reading j, $f_r = 1$ when r finds no changed or missing requirement, and $c(F) = 1$ when all code checks pass. The set is kept when

$$\mathrm{keep}(F) = c(F) \wedge \bigwedge_r \left[f_r \wedge H^{(r)}_{\{A,B\},\{A,B\}} = I_2 \right] \wedge \bigvee_r \left[H^{(r)}_{\circ A} \wedge H^{(r)}_{\circ B} \right]. \quad (3)$$

B LIBRARY COMPOSITION

Table S2 gives the labels and counts in the fixed example library for every type. "Open" includes both unresolved and contested records. "Settled" denotes failure records the reviewers label as settled by the wording; this label is not the word-list check used for the defaults experiment. Failures that differ only in output format (51 with two accepted readings, 10 with one) are not choices of meaning and are not in the table. Possible reference errors (27) and unplaced cases (four) are also kept apart.

Table S2 counts the 204 accepted request sets by type, 43 of them with a choice in a filter (Appendix C). Over the 215 sets whose choice a model proposes, aggregation level contributes 63, ties 44, missing values 40, counting unit 35, calculation 15, relationship role nine, and time scope two. The remaining seven belong to the six additional labels: running-total interpretation (two), active-status filtering, comparison baseline, baseline population, baseline definition, and row-count versus quantity measurement (one each). These counts reflect the generation and selection procedure. For

Algorithm S1 Build a request set around one calculation choice

Require: Databases D, target depths d, required operations R
Ensure: Accepted request sets and reasons for rejected sets

for each requested task **do**
  Sample $p_A \in \mathcal{P}(D, d, R)$ with the look-ahead of Eq. (2)
  If no query is delivered, record the shortfall and continue
  Describe p_A; write and check its request
  Propose edits $p_A \to p_B$ that express a different choice
  Execute and describe both queries; require different answers
  Repair failed proposals once; review both readings
end for
Select one choice per task at most, balancing types and databases
for each selected choice **do**
  List what stays fixed and what changes
  Write requests specifying A, specifying B, and leaving it open
  Check requirements and all six request–reading pairs
  Revise a failing set once and recheck
  Keep F if keep(F); otherwise record the reason
end for

Table S1: Program construction on 12 databases, with no model calls after catalogue preparation. Every delivered SQL is distinct, has the requested depth, and passes both integrity checks.

Source	DBs	Delivered	Constructions	Seconds
Spider	4	398/400	289	127
BIRD	4	400/400	270	274
LiveSQLBench	4	400/400	309	37
Total	12	1198/1200	868	437

Source-level times are rounded individually; the total is rounded from the unrounded sum. Separate task review delivers 384/400 tasks, and language generation supplies 382/384 requests.

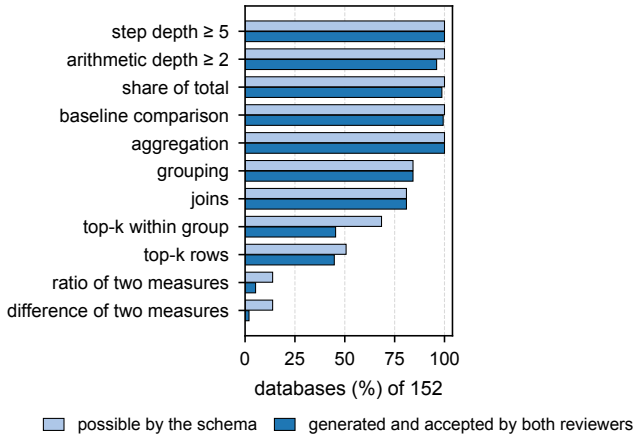


Figure S1: Features the generator can build, as a share of 152 databases: possible by the schema, and generated and accepted.

Table S2: All choice types (Figure 4 of the paper shows the major ones). Library: choices of meaning in benchmark failures, open (the wording fits both readings) or settled (it selects one). References: BIRD and Spider questions silent on the choice, by the reading their reference SQL takes: the usual one, or the other with its share (%), after the less common readings were reviewed; –: not readable from the SQL. Sets: the 204 accepted request sets, of the types whose other reading at least 1% of silent references take. The library's value-representation type is split by its SQL edit: comparison strictness is a strict cutoff, any other edit value matching.

		Library		References		
Choice type	The choice	Open	Settled	Usual	Other (%)	Sets
Aggregation level	Rows or group averages?	6	2	244	15 (5.8)	61
Calculation	Which formula is meant?	1	1	613	1 (0.2)	0
Quantifier	"Than any": one or all?	3	0	8	7 (46.7)	0
Condition linkage	Conditions on one row?	1	0	260	0 (0.0)	0
Counting unit	Count rows or entities?	7	27	2,521	237 (8.6)	30
Missing values	Keep or skip missing?	8	3	1,630	26 (1.6)	29
Negation	Some non-match, or none?	0	1	272	3 (1.1)	1
Population scope	Keep unmatched rows?	0	1	12,462	5 (0.0)	0
Referent column	Which column is meant?	19	6	–	–	0
Relationship role	Which link to follow?	2	4	15	0 (0.0)	0
Strict cutoff	Strict or inclusive?	1	0	309	5 (1.6)	42
Ties and selection	One tied row, or all?	7	3	2,826	397 (12.3)	41
Time scope	Period? Ends included?	1	2	254	1 (0.4)	0
Value matching	Exact or loose match?	1	3	9,330	22 (0.2)	0
Total		57	53			204

Table 1, AmbiQT's column and table ambiguity and AMBROSIA's vagueness map to *which column*, AMBROSIA's scope to *any vs. all*, and attachment to *how conditions link*.

The References columns of Table S2 read each reference query with an SQL parser and the schema's keys and column types, without a model, and count a question only when its wording has no cue for the choice. Ties, counting unit, missing values, population scope, "between" and value matching follow Section 6.1 of the paper; calculation is a question that says percent, answered times 100 or as a fraction. *Quantifier*: a question that says "than any", answered by comparing with every value (`> MAX`, `< MIN`, `ALL`) or with at least one (`> MIN`, `< MAX`, `ANY`). *Negation*: a negated question answered with no matching record (`NOT IN`, `NOT EXISTS` or `EXCEPT` over another table) or with some non-matching record (<> on the many side of a join). *Aggregation level*: an entity ranked by a numeric column on the many side of a join, by one row's value or by a sum or average per group. *Condition linkage*: two or more conditions on different columns of the many side of a join, on one row or on separate rows (the table joined twice by the same key, or an `INTERSECT`). *Relationship role*: two tables that share a parent and are also connected by a link table, joined through the parent or through the link table. Which column a phrase means depends on the words, not the SQL, so the referent column has no count. Each rule reads one SQL form of its choice, so it detects only part of the choices mined from failures: applied to the 85 library records of the twelve types, the rules find the choice in 50, including all ties and quantifier records, 27 of 34 counting records and 4 of 6 relationship records, but none of the aggregation, time scope or value matching records, whose forms (grouping by a non-key attribute, an average used as a threshold) the rules do not read. The counts are therefore a lower bound on how often the choices arise. Every reference that takes the less common reading was then read against fixed criteria (the SQL takes that reading, the wording leaves the choice open, and the other reading could change the answer), and every verdict is listed for the author's review (`human_supervision/review-reference-minority.md`). Of 1,259 such references, 540 are dropped, most often because the wording settles the choice ("list the students with the highest", "how many different"), the join cannot repeat the counted entity, or a `LIKE` only escapes an apostrophe. Only this side was reviewed, so the shares in the table are conservative. The same drops apply to Section 6.1 of the paper.

C REQUEST SETS WITH FILTER CONDITIONS

The filter sets of Section 2.4 of the paper come from one optional rule of the generator. It adds a single condition to the first stage of a query, on the rows of that stage's table, and builds the second reading at the same time. The five forms are: a date range, inclusive (>= and <=) or exclusive (> and <), with ends at the 30th and 70th percentile of a date column; a cutoff at a frequent value of a measure, `>= v` or `> v`; "without" a category, no record of the entity has it (`NOT IN`) or some record lacks it (`IN` with <>); two categories, on one record or on any records of the entity; and a per-entity count, over entities with matching rows (inner join) or over all entities (outer join, zero counts kept). A condition is used only if it keeps 20–80% of the stage's rows, and the generator prefers forms it has used less often. The seed is 20261016; 161 of 180 requested queries are delivered, 121 of them have readings with different answers, and 92 request sets pass all checks.

Table S3 gives the results per model on the 43 accepted filter sets, and Figure S2 by choice type. The author rejects all 17 date-range sets. Their open requests say "between," which SQL's `BETWEEN` includes, and so do 254 of the 255 BIRD and Spider references whose question says "between" with no other cue; the wording therefore settles the ends. The one exception is a Spider reference that applies strict bounds to the start and end columns of a staff role.

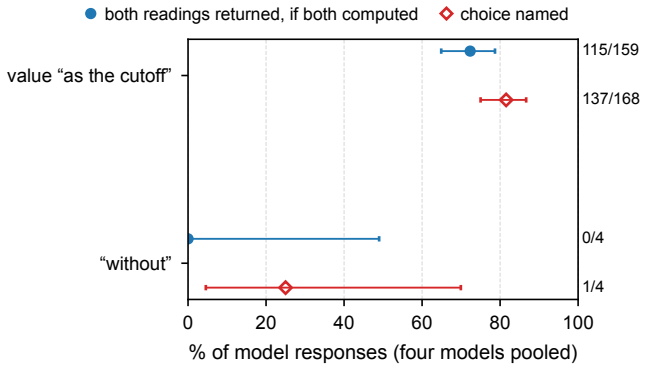


Figure S2: Filter request sets (43), four models pooled, by the choice in the filter and the wording of the open request (Wilson 95% intervals).

Table S3: Filter request sets (43 accepted). Top, per model: both specific requests right (with exploration); among those, open requests answered with both readings; open requests that name the choice (Opus judge); specific requests right (single call); missed readings the model computes when asked, of all missed.

Figure S2 gives the results by choice type.

Model	Both right	Both returned	Named	Right	Missed, computable
Haiku	42	33	40	40	9 / 10
Sonnet	41	36	38	42	5 / 6
GPT-Sol	42	32	35	40	10 / 10
GPT-Luna	38	14	25	41	24 / 29

A BIRD reference whose evidence says `BETWEEN 121 AND 130` but whose query excludes both ends is not counted, since its evidence settles the ends. Four further references use strict bounds moved one step out, or a negated range, and select the inclusive range. What models raise follows the wording of the open request. Forty of the 42 cutoff requests give a value “as the cutoff” or “threshold,” with no comparison word, and the other two say “upwards of”; models return both the strict and the inclusive reading in 115 of the 159 cases where they compute both.

The settled request is never called open. GPT-Luna returns both readings least often, as on the other sets. Of the 49 rejected sets, 17 are the date ranges, 15 are cutoffs whose wording (“passes”, “clears”, “on the high side of”) already makes the comparison strict, five are “without” requests that already mean no such record, and one is a per-entity average whose request restricts the population. The other 11 test per-entity counts with or without unmatched entities (eight) and conditions on one or any records (three), choices whose other reading real references almost never take (Figure 4 of the paper).

D CALCULATION CHOICES WITH REAL CONSEQUENCES

Table S4 surveys documented cases, across six domains, in which one open or mistaken choice of the kind our library records changed a consequential number or decision. None is a text-to-SQL failure. Each entry was checked against the original paper or official report; we left out cases whose primary source we could not reach, such as a bank’s value-at-risk spreadsheet and a classic study of graduate admissions.

E FURTHER DETAILS

Seeds, versions and reruns. The grammar is `compose-v3.4`. Task requests, Spider splitting, request set selection, explanation allocation, and bootstrap resampling use seed 20261003; BIRD hold-out selection uses 20261001; the paired wording comparison uses 20261004. Coverage uses seed 1 and depth-control runs seeds 1–3. Python was 3.11.5 and SQLite 3.41.2. During the experiments, the Claude command-line client changed from 2.1.287 to 2.1.289 and the Codex client from 0.147 to 0.160. Temporary usage-limit outputs were removed and retried rather than treated as SQL answers. A final query that failed under concurrent execution was rerun alone: ten such answers then matched their reference (nine GPT-Luna, one GPT-Sol). Token counts are unavailable for the explanation runs, so latency is reported instead of cost.

Rewrite comparison. The comparison of the original specific request (a) with the other texts was fixed before the run. The comparisons with the edited request (c) were added after an interim look at half of the readings.

Response selection. The explanation examples are the first saved responses displaying each tested pattern in file order. The worked aggregation example uses the smallest case-identifier hash among eligible request sets. Library examples use the first eligible record per type at high confidence, or medium confidence if none is high, with fixed benchmark and task ordering. Generated examples outside library labels enumerate retained cases before selecting the four shown in the main text. These selection rules support traceability; illustrative examples are not frequency estimates.

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Table S4: Documented cases where one calculation choice changed a consequential number or decision, by domain, with the closest choice type in our library.

Domain	Case	Choice and consequence	Library type
Economics and policy	Public debt and growth [3]	Growth was averaged within each country before averaging across countries, so one year of New Zealand data weighed as much as decades of another country. With corrected data and weighting, high-debt countries grew 2.2% a year, not −0.1%.	aggregation level
Health care	Risk algorithm for extra care [6]	"Need" was computed as health-care cost. The algorithm looked accurate by standard measures, yet fixing the proxy would raise the share of Black patients receiving extra help from 17.7% to 46.5%.	which column a phrase means
Health care	Hospital mortality rates [7]	Four methods, given the same 2.5 million discharges, included 28–95% of them. 12 of 28 hospitals with higher-than-expected mortality by one method had lower-than-expected mortality by another.	population scope
Public health	1995 UK contraceptive pill warning [2]	A warning of higher thrombosis risk for two pill types was followed by pill use among under-16s falling from 40% to 27%, and by estimated NHS costs of £21 million for maternity care and £46 million for abortions. The author concludes that risk should be more carefully presented.	how a quantity is stated (not in our library)
Social science	Referees and skin tone [8]	29 teams analysed the same data. Their odds ratios ranged from 0.89 to 2.93; 20 teams found a significant effect and 9 did not, using 21 combinations of covariates.	several analysis choices
Science	Brain imaging [1]	70 teams analysed one dataset to test the same 9 hypotheses. No two chose the same workflow, and the results of the hypothesis tests varied substantially.	several analysis choices
Science	Gene names in spreadsheets [9]	Spreadsheet software converts gene names to dates and numbers by default; about one fifth of papers with supplementary gene lists in spreadsheets contain such conversions.	value representation
Engineering	Mars Climate Orbiter [5]	Thruster data in English units were used where the trajectory software expected metric units; the spacecraft was lost.	value representation
Engineering	Ariane 5 flight 501 [4]	A 64-bit floating-point value was converted to a 16-bit signed integer that could not hold it; the launcher broke up about 40 seconds after lift-off.	value representation